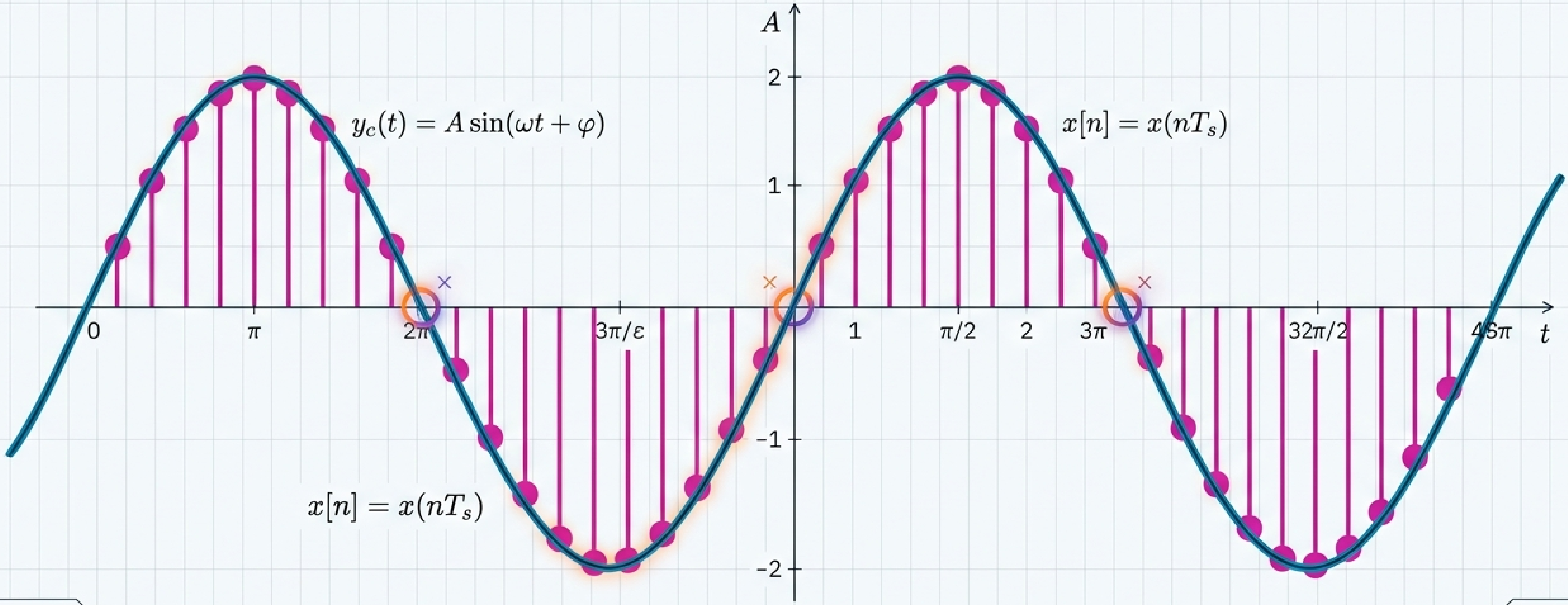
CHAPTER 1

Generalities on Signals

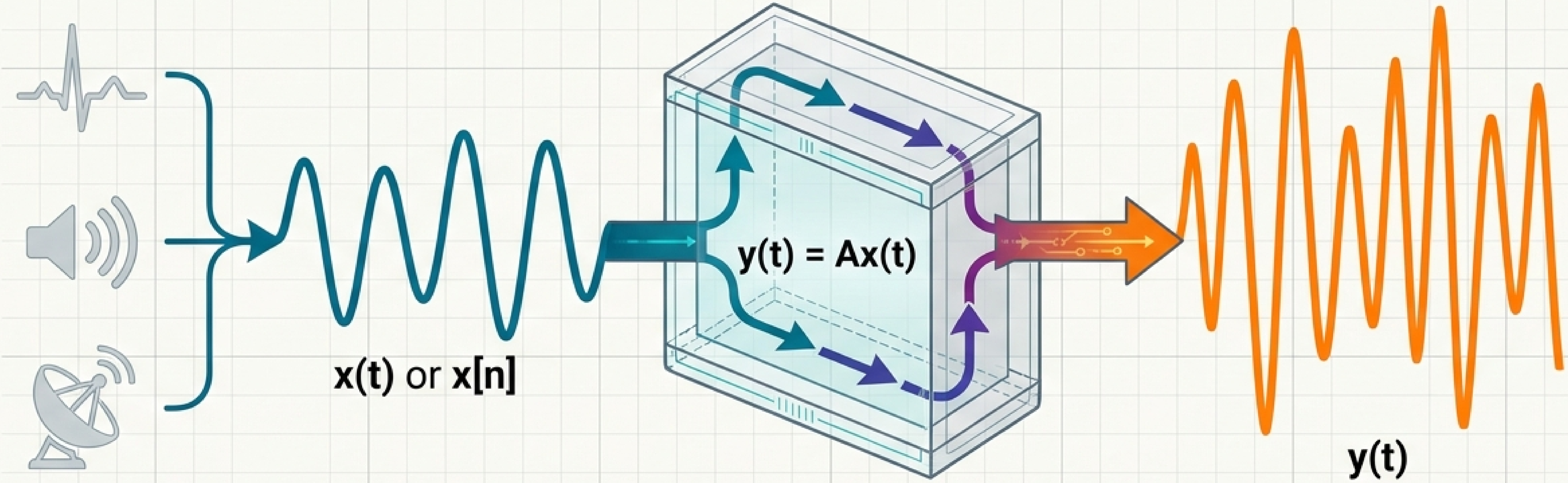
The mathematical foundation of signal processing.

x[n] =	0.61
x[n] =	0.83
x[n] =	0.22
x[n] =	0.61



A..	0.0
REV	0.0
NEV	0.0%5

METRICS



The Signal

A mathematical quantity carrying information about a physical phenomenon.

The System

The operation or physical device that transforms the input signal into an output signal.

The Goal

To represent, analyze, transform, transmit, store, and extract information.

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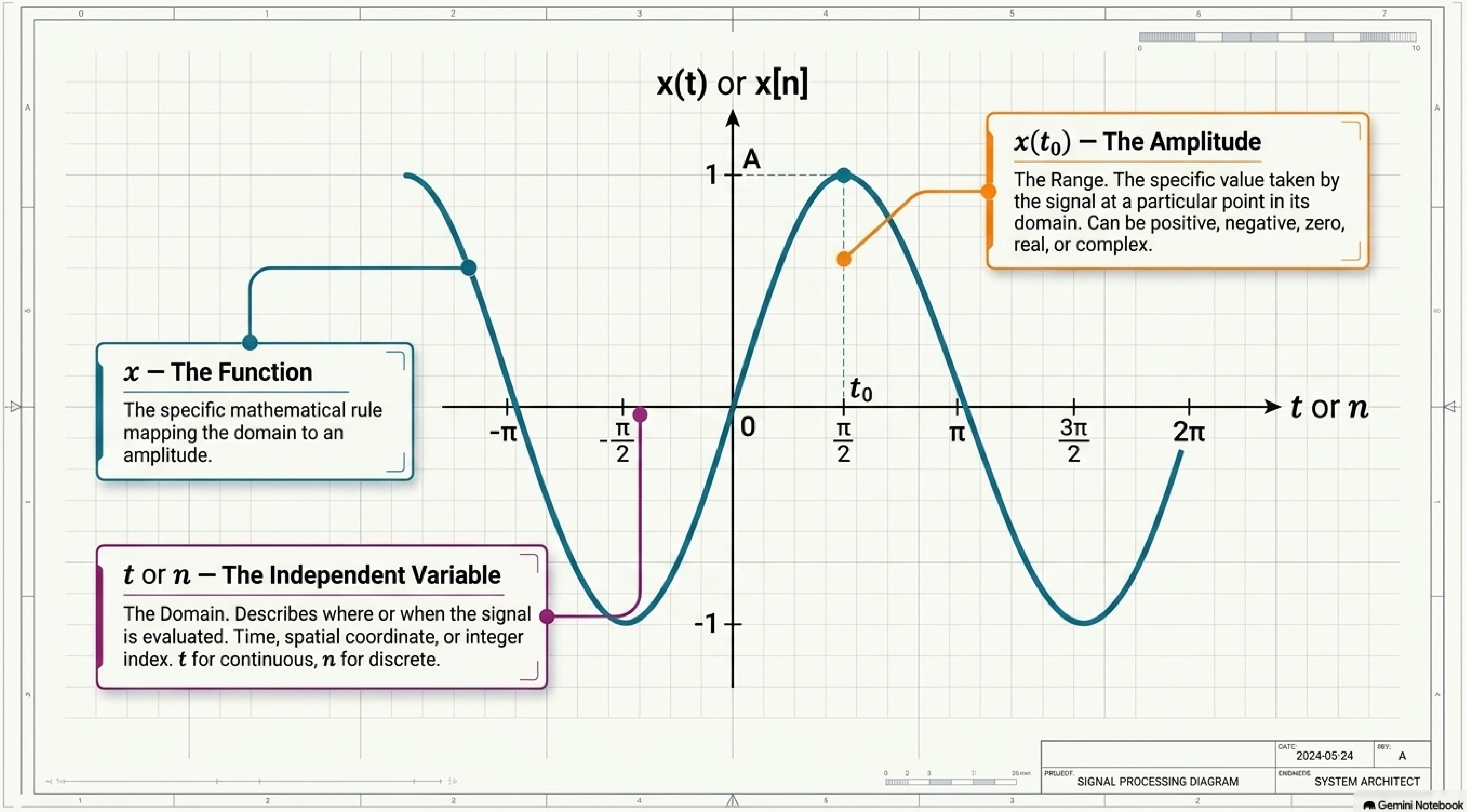

x(t) or x[n]
A
1
-1
t or n
-π
-π/2
0
π/2
π
3π/2
2π
t_0
$x(t_0)$ — The Amplitude
The Range. The specific value taken by the signal at a particular point in its domain. Can be positive, negative, zero, real, or complex.
x — The Function
The specific mathematical rule mapping the domain to an amplitude.
t or n — The Independent Variable
The Domain. Describes where or when the signal is evaluated. Time, spatial coordinate, or integer index. t for continuous, n for discrete.
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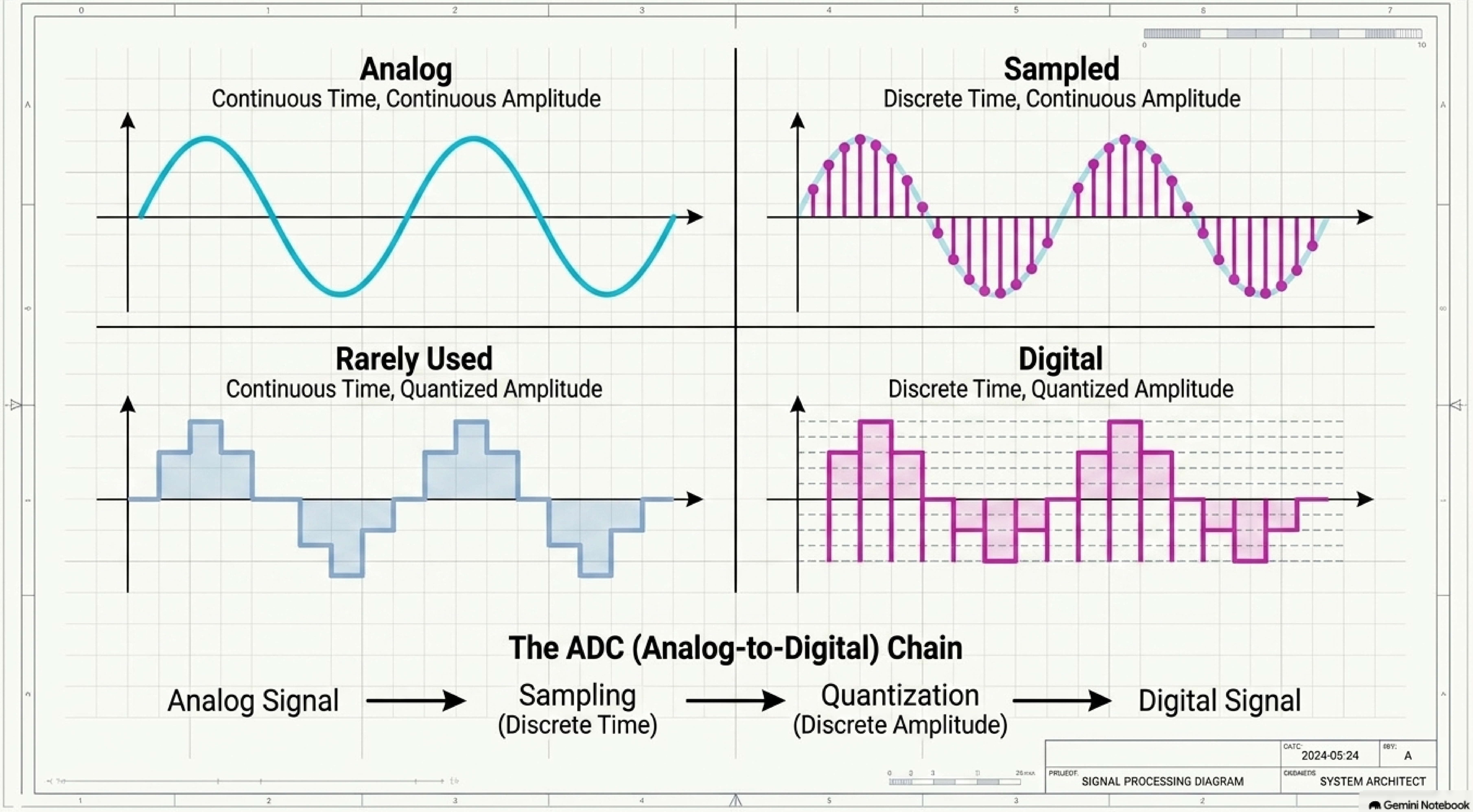

Analog
Continuous Time, Continuous Amplitude
Sampled
Discrete Time, Continuous Amplitude
Rarely Used
Continuous Time, Quantized Amplitude
Digital
Discrete Time, Quantized Amplitude
The ADC (Analog-to-Digital) Chain
Analog Signal
Sampling
(Discrete Time)
Quantization
(Discrete Amplitude)
Digital Signal
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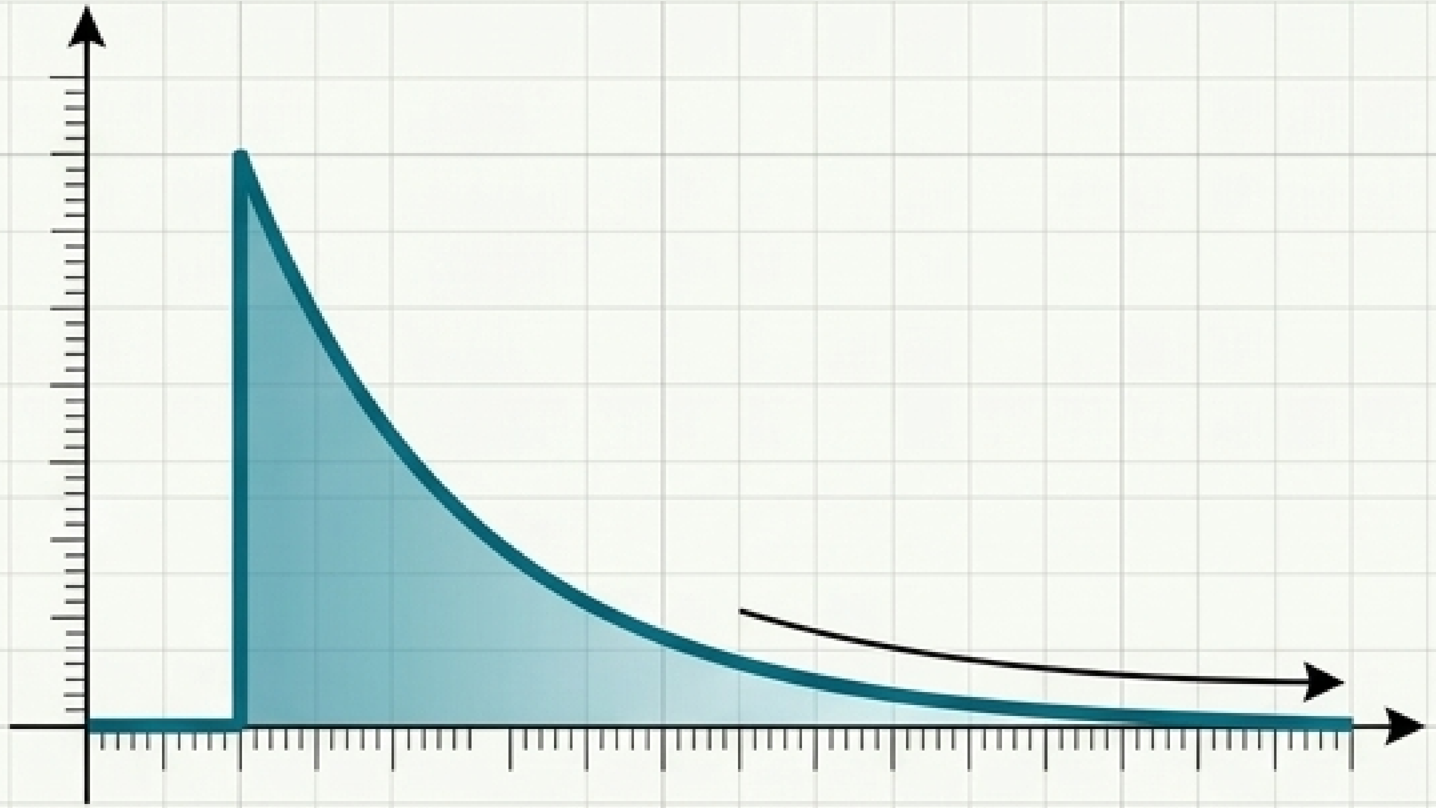
Signal DNA Dashboard

Predictability	**Deterministic** Exact mathematical specification (e.g., $x(t)$=3cos(100pi t)`)	Random (Stochastic) Described statistically (e.g., thermal noise)
Time Direction	**Causal** Zero before `$t$=0`; system cannot respond before input occurs	Non-causal Nonzero for `$t < 0$`
Time Support	Infinite-duration Endlessly continuing (e.g., decaying exponential)	**Finite-duration** Nonzero only over a finite interval
Values	**Real-valued** $x(t)$ is a real number	Complex-valued $x(t) = x_R(t) + jx_I(t)$

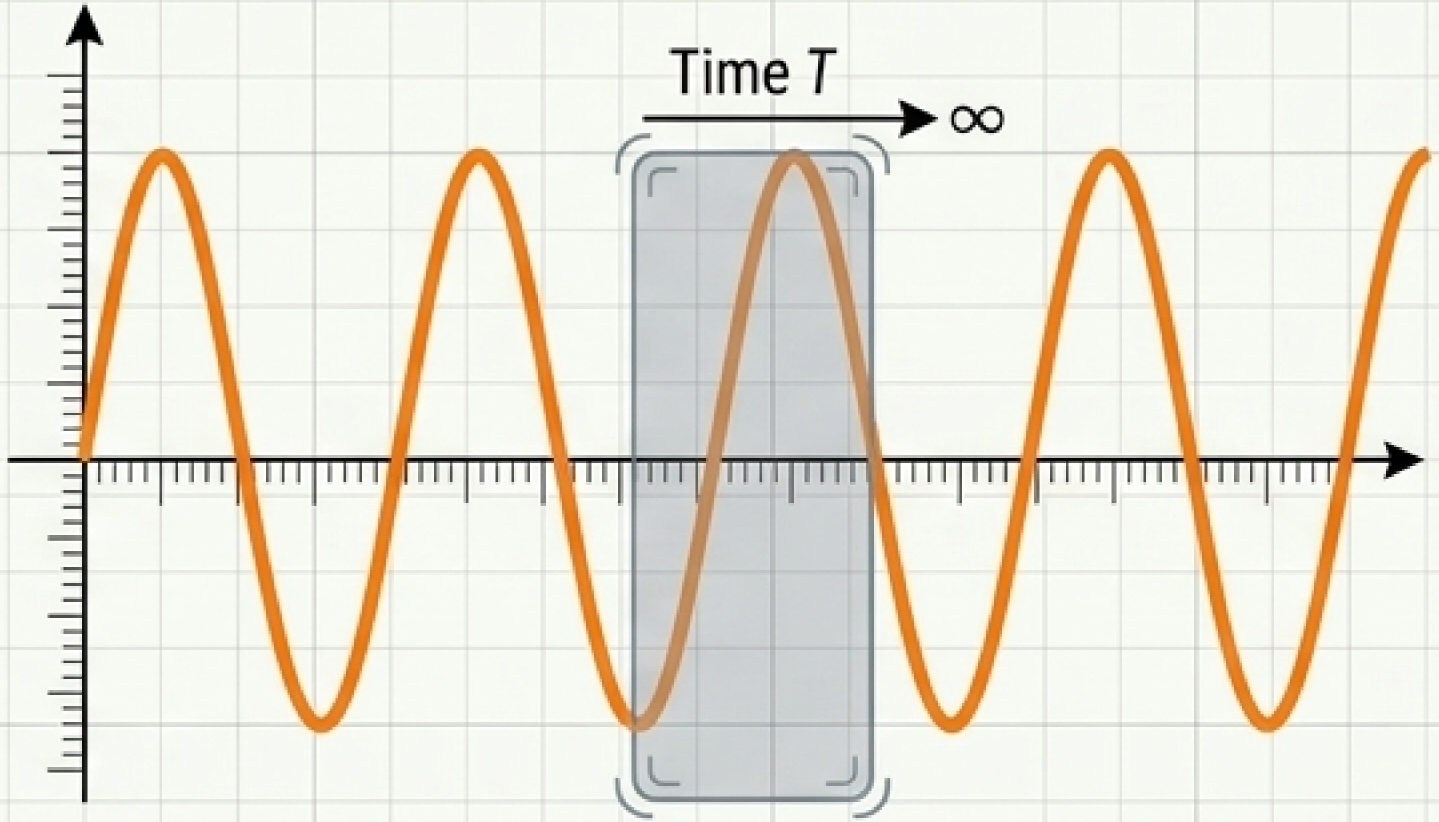
These properties are independent. A signal can be categorized across all these dimensions simultaneously.

Energy Signals



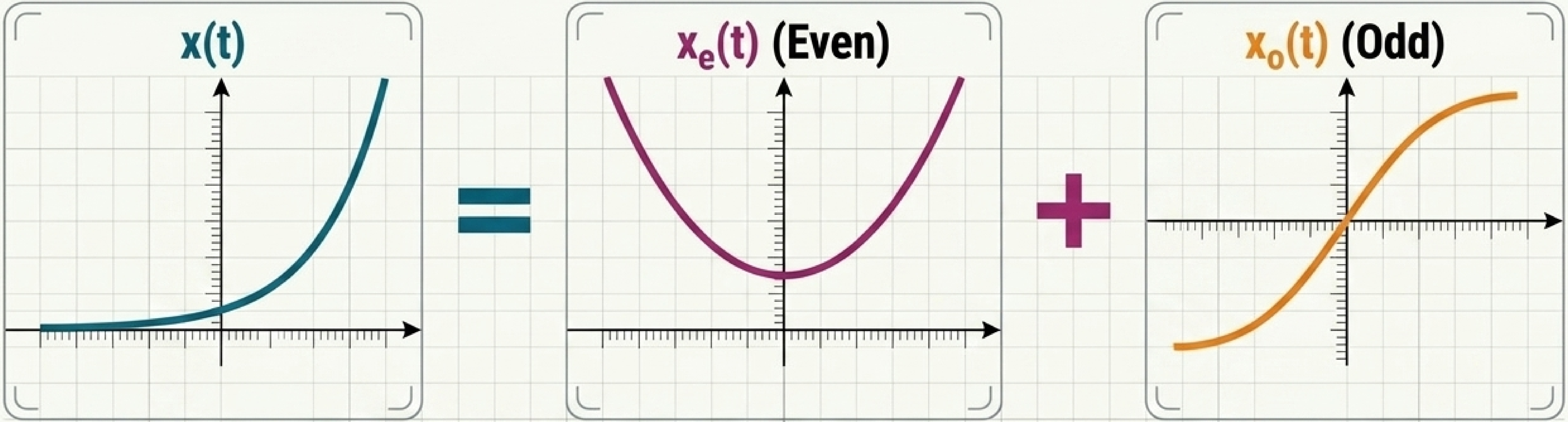
Definition:	$0 < E_x < \infty$
Math:	$E_x = \int \lvert x(t) \rvert^2 \, dt$
Profile:	Finite-duration or decaying signals. Average power is exactly zero.

Power Signals



Definition:	$0 < P_x < \infty$
Math:	$P_x = \lim_{T \to \infty} \frac{1}{2T} \int \lvert x(t) \rvert^2 \, dt$
Profile:	Infinite-duration, periodic signals. Total energy is infinite.

Key Rule: A nonzero signal cannot be both. It is either an energy signal, a power signal, or neither.



Even (Symmetric)
$x(-t) = x(t)$
Symmetric around the vertical axis.
Calculated via: $x_e(t) = 0.5[x(t) + x(-t)]$

Odd (Anti-symmetric)
$x(-t) = -x(t)$
Asymmetric. Must pass through $x(0) = 0$.
Calculated via: $x_o(t) = 0.5[x(t) - x(-t)]$

The Universal Rule: Any signal can be decomposed into an even and odd part.

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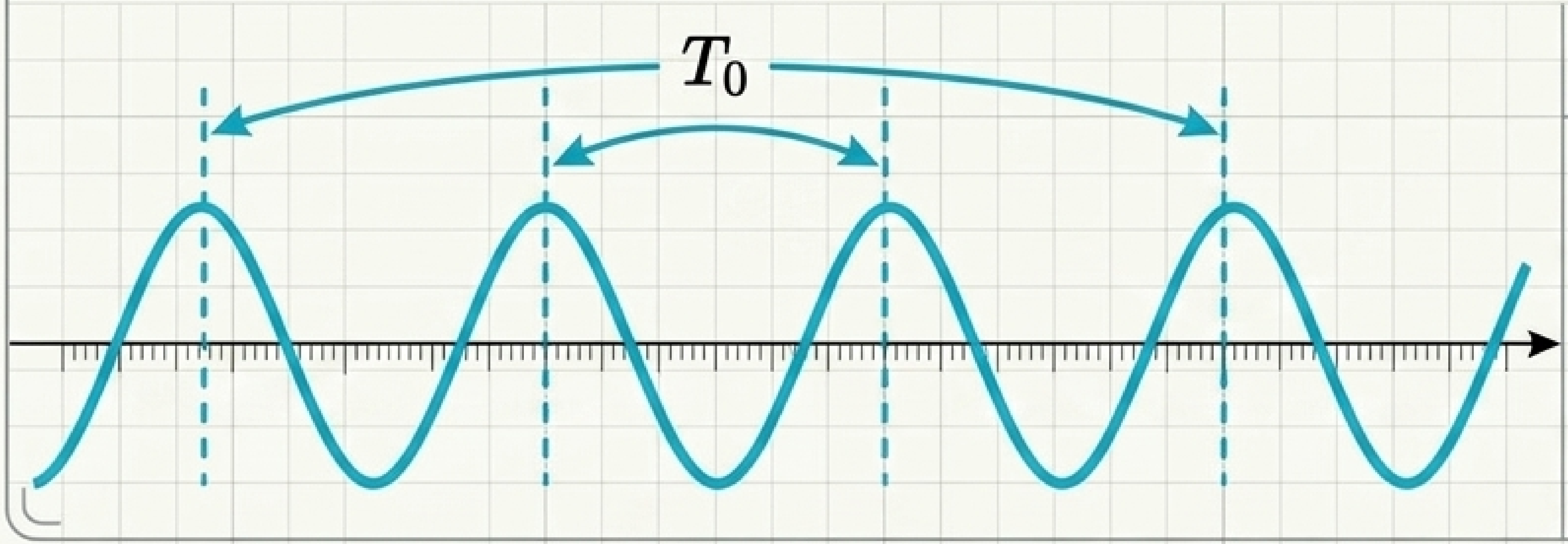
Signal Periodicity Paradox

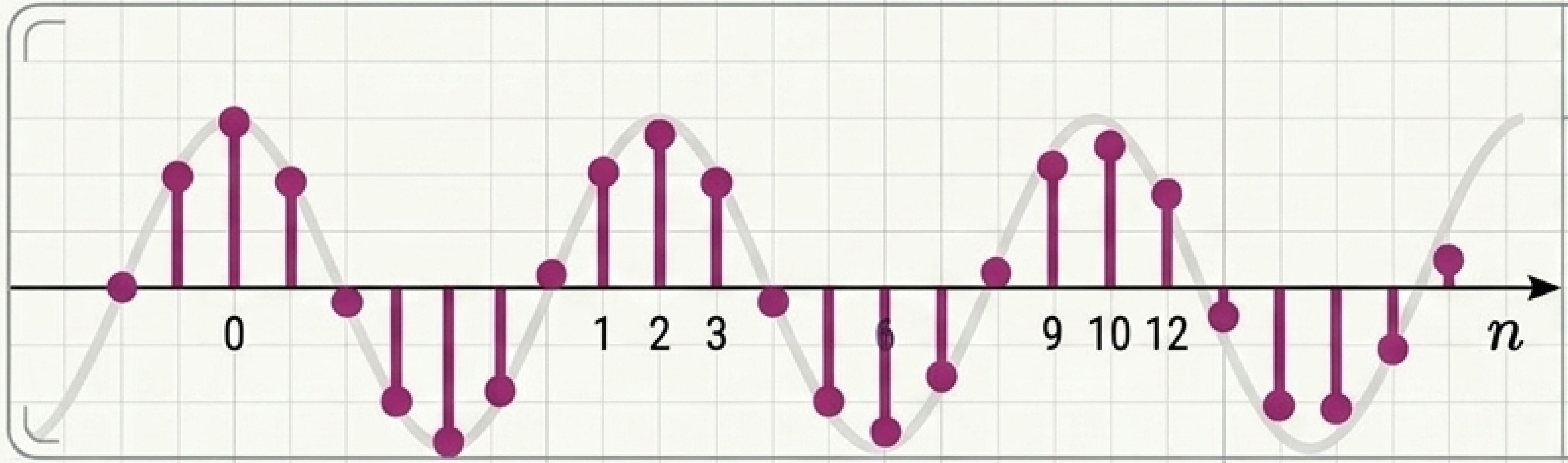


Continuous-Time: Always Periodic

$$x(t + T_0) = x(t)$$

for some $T_0 > 0$.

This signal tiws aligns the T_0 to period phase drift repeated peoificity on repetition, and soration of am full cycle.



Discrete-Time: The Periodicity Paradox

$$x[n] = \cos(\sqrt{2}\pi i\, n)$$

Only periodic if the normalized frequency is a rational number. An irrational frequency prevents the integer index from ever perfectly aligning with a full cycle.

A **continuous sinusoid** always repeats. A discrete sinusoid with an irrational frequency **NEVER** perfectly repeats.

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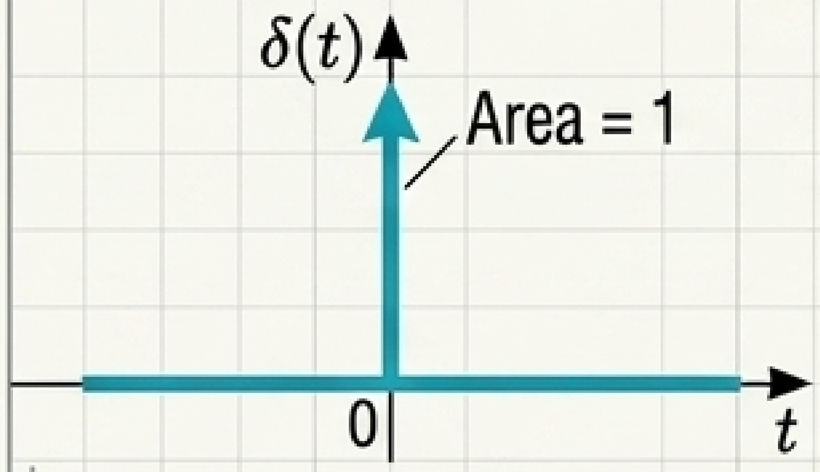
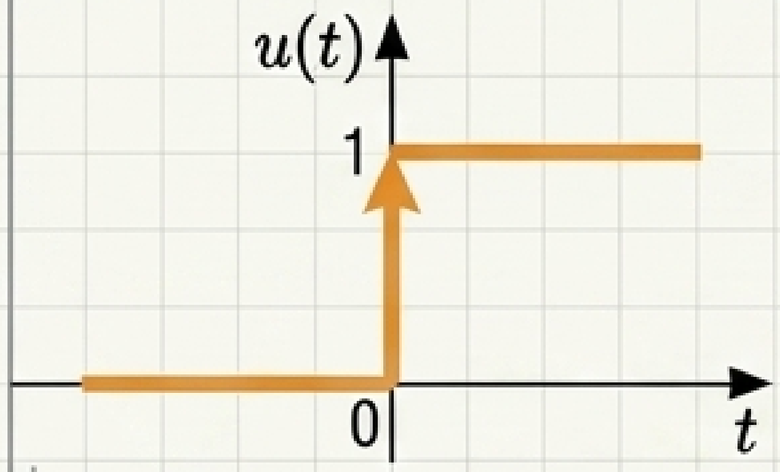
The Standard Vocabulary: Foundations

Unit Impulse $delta(t)$



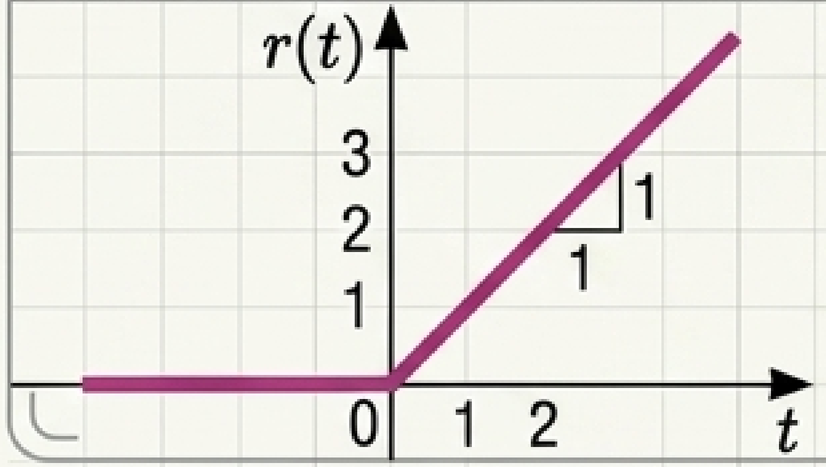
"The Infinite Spike."
$Area = 1$, zero everywhere else. Fundamental for sampling, convolution, and sifting.

Unit Step $u(t)$



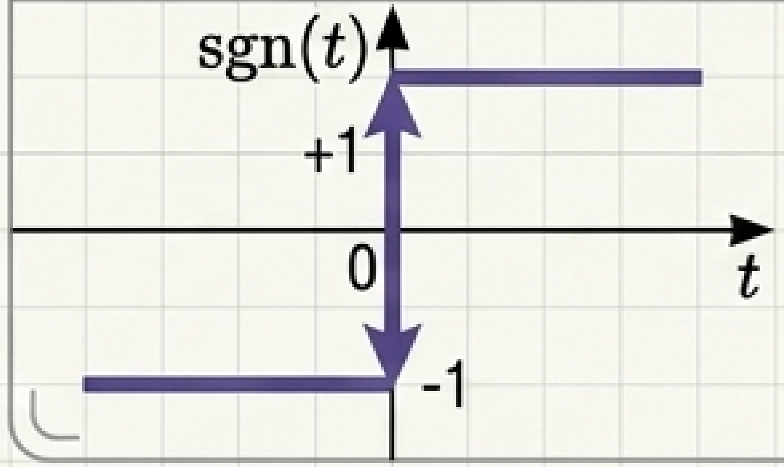
"The Switch."
0 before $t = 0$, 1 after.

Ramp $r(t)$



"Linear Growth."
Defined as $t * u(t)$. Models linearly increasing quantities.

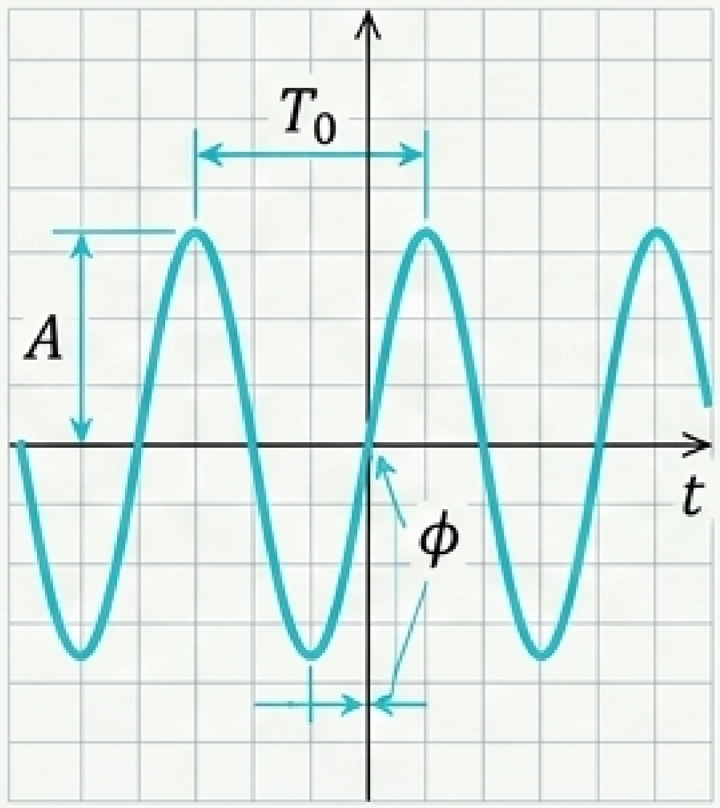
Signum $\mathrm{sgn}(t)$



"The Polarity Indicator."
Odd signal, related to the step: $\mathrm{sgn}(t) = 2u(t) - 1$.

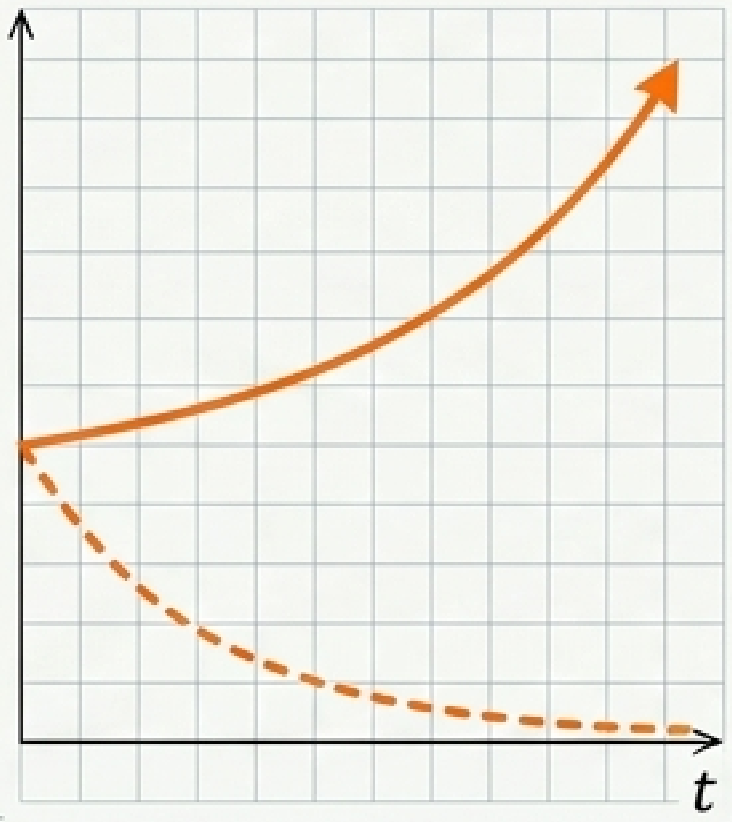
The Standard Vocabulary: Modulators & Pulses

Sinusoid



$$A\cos(\omega_0 t + \phi)$$

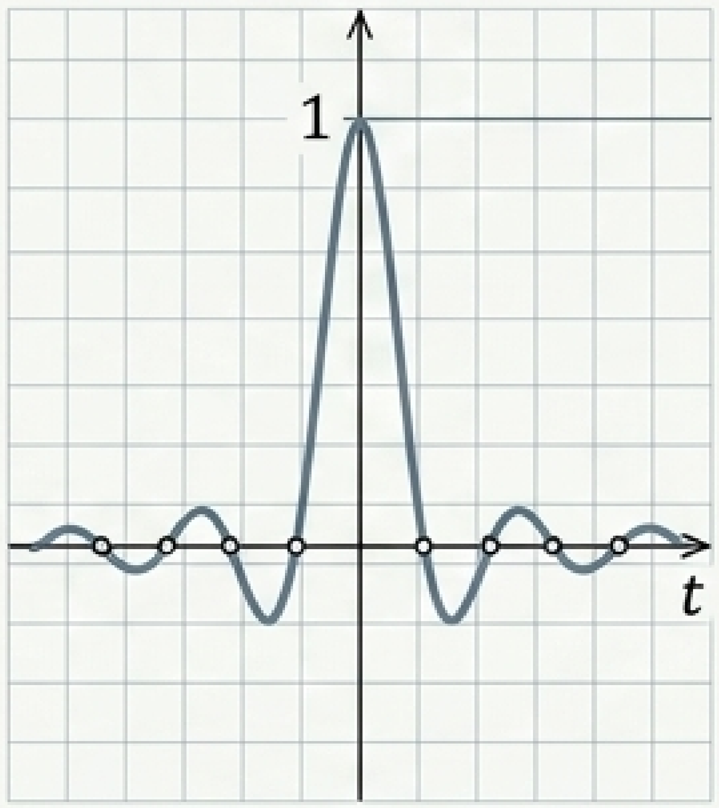
Exponential



$$A\,e^{at}$$

Notes Euler's relation for complex exponentials: $e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$.

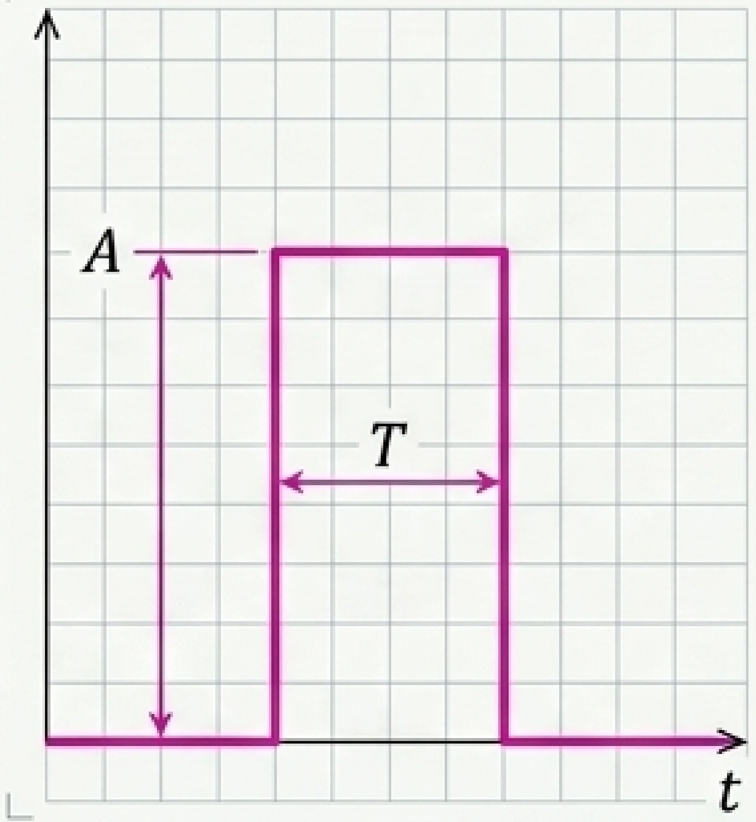
Sinc



$$\frac{\sin(\pi t)}{\pi t}$$

"The Ideal Filter." Fundamental in sampling theory.

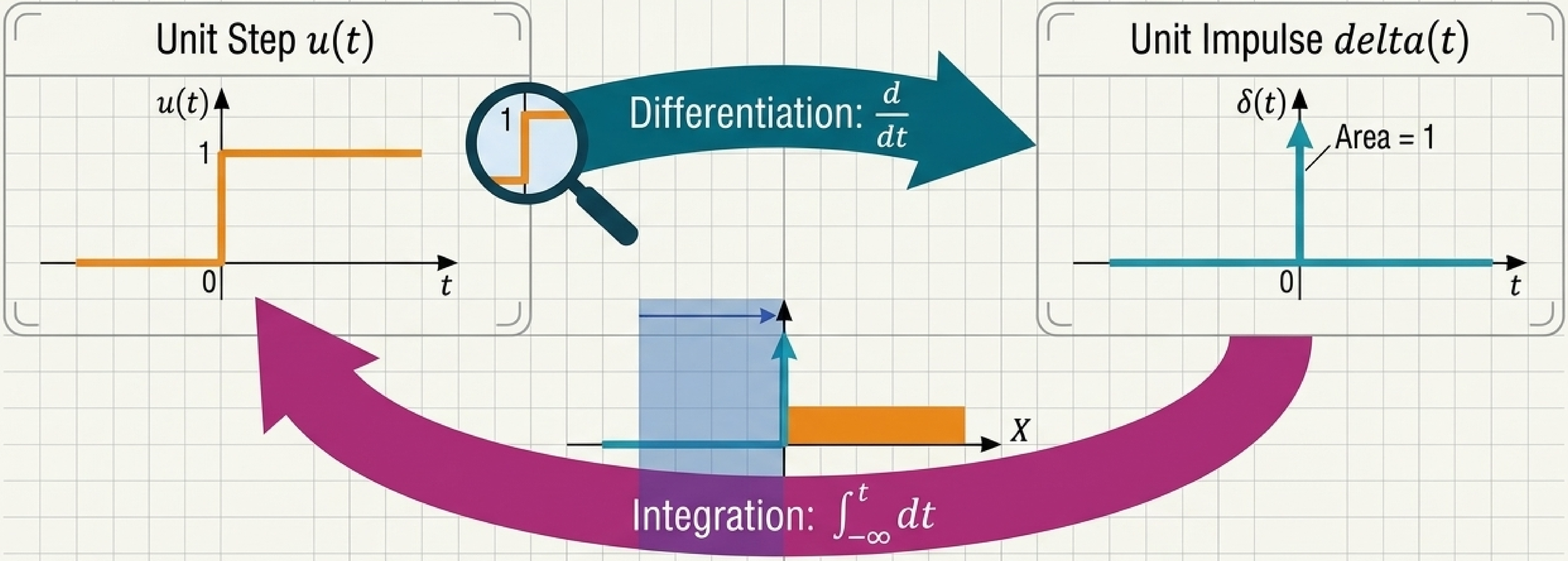
Rectangular Pulse



$$\mathrm{rect}(t)$$

"The Window." Built from combined unit steps.

The Calculus of Signals: Visualizing Differentiation and Integration



The infinite spike is the instantaneous rate of change of the switch.
The switch is the accumulated area of the spike.

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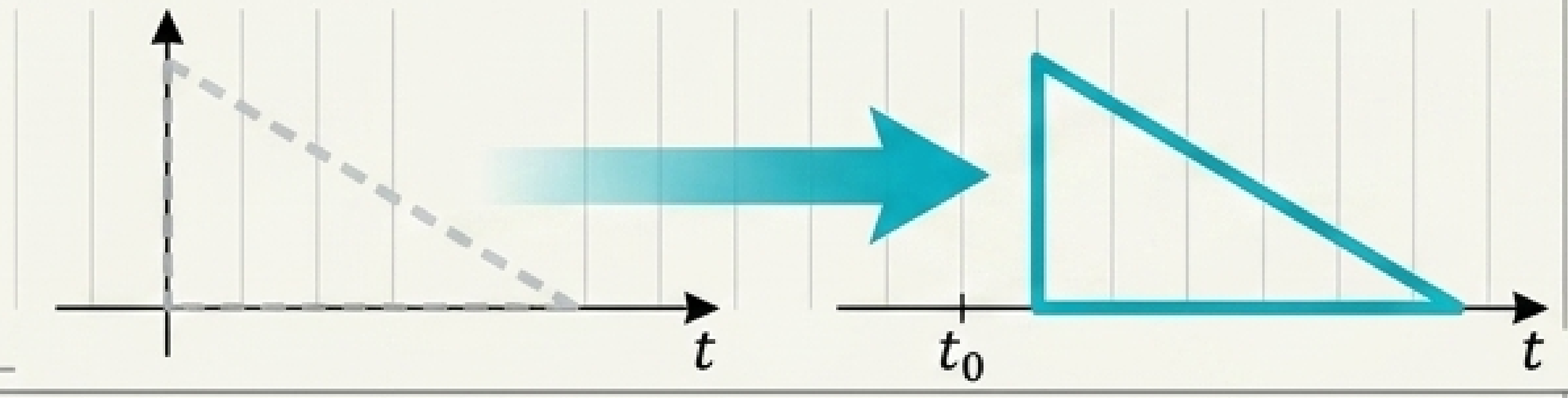
Time Transformations

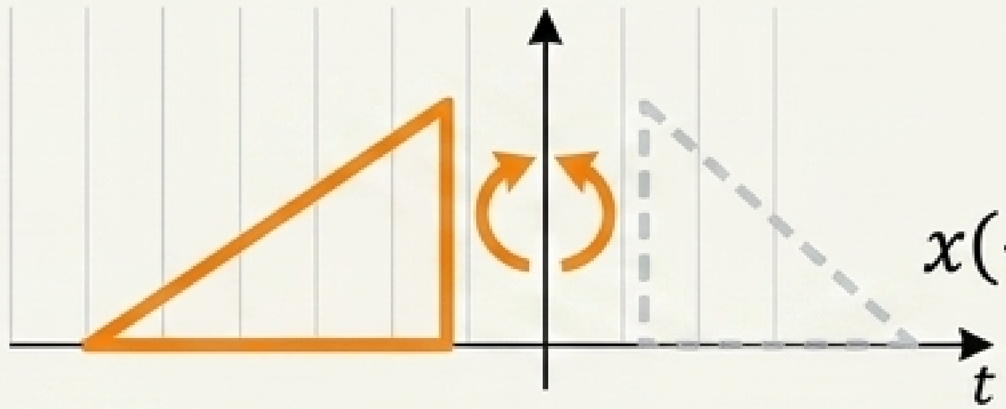
Shift



$$x(t - t_0)$$

A negative sign ($-t_0$) means a rightward Delay.
A positive sign ($+t_0$) means a leftward Advance.
This is the most common beginner error.

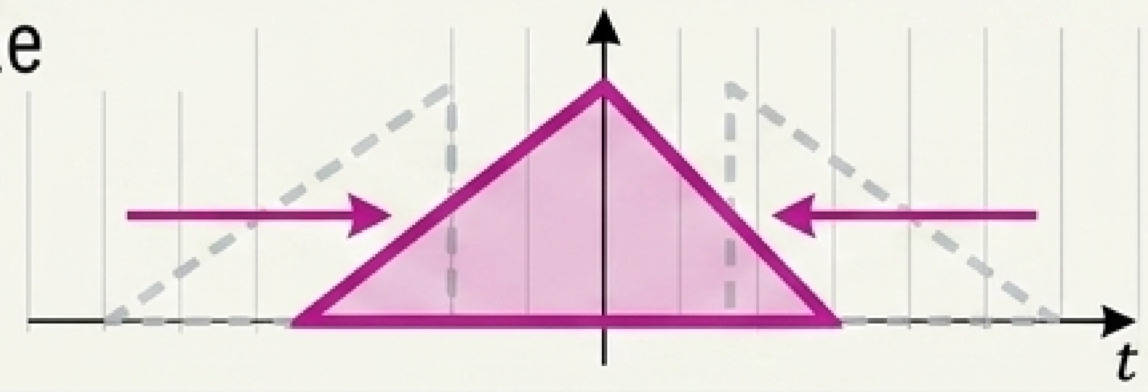
Reverse



$$x(-t)$$

$x(-t)$ mirrors perfectly across the vertical axis.

Scale

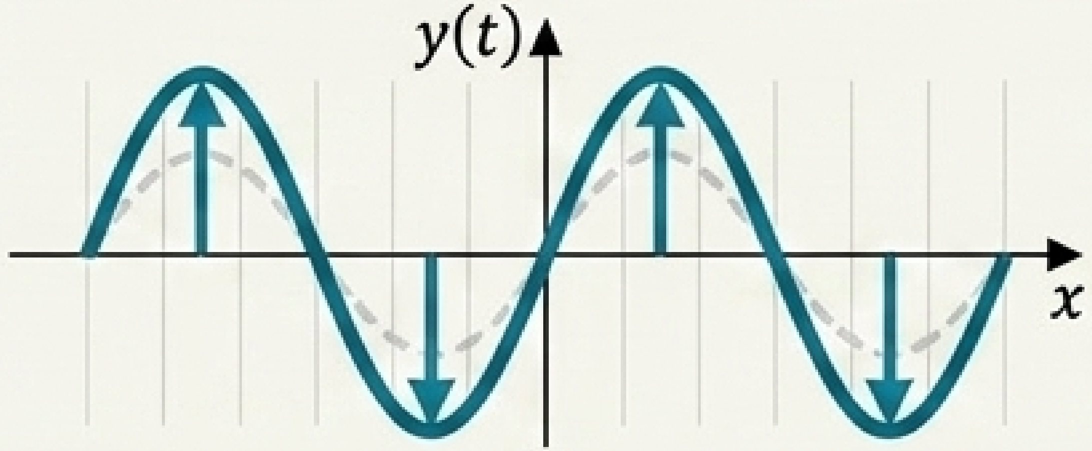


$$x(at)$$

- If $a > 1$, horizontal compression.
- If $0 < a < 1$, horizontal expansion.

Amplitude Transformations

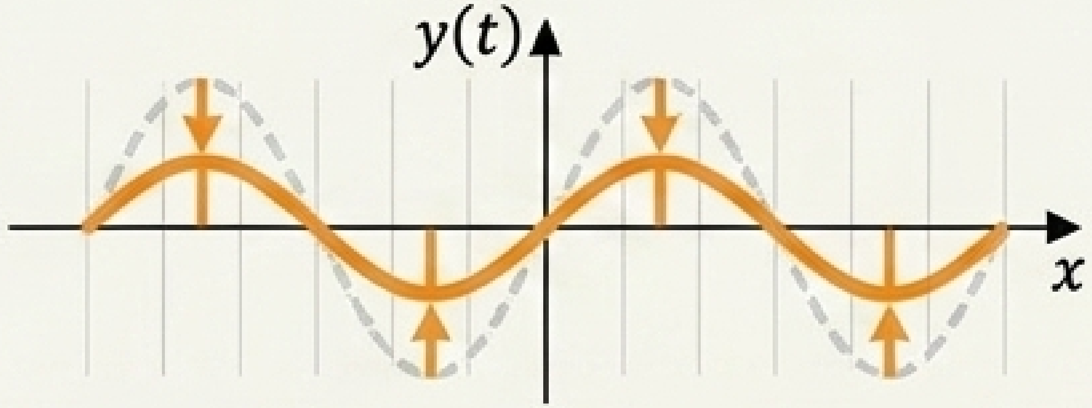
Amplification ($A > 1$)



$$y(t) = Ax(t)$$

$A > 1$ results in vertical stretching.

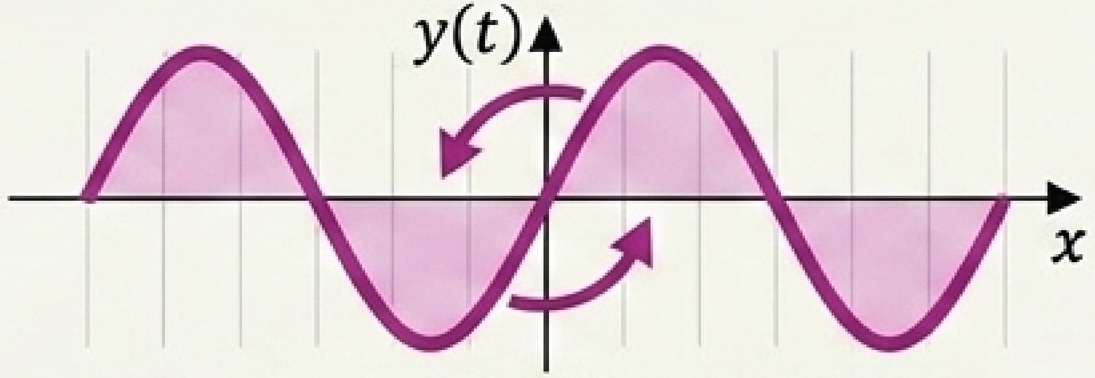
Attenuation ($0 < A < 1$)



$$y(t) = Ax(t)$$

$0 < A < 1$ results in vertical compression.

Inversion ($A < 0$)



$$y(t) = -x(t)$$

$A < 0$ results in reflection across the horizontal axis.

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Synthesis: Deconstructing a Complex Signal

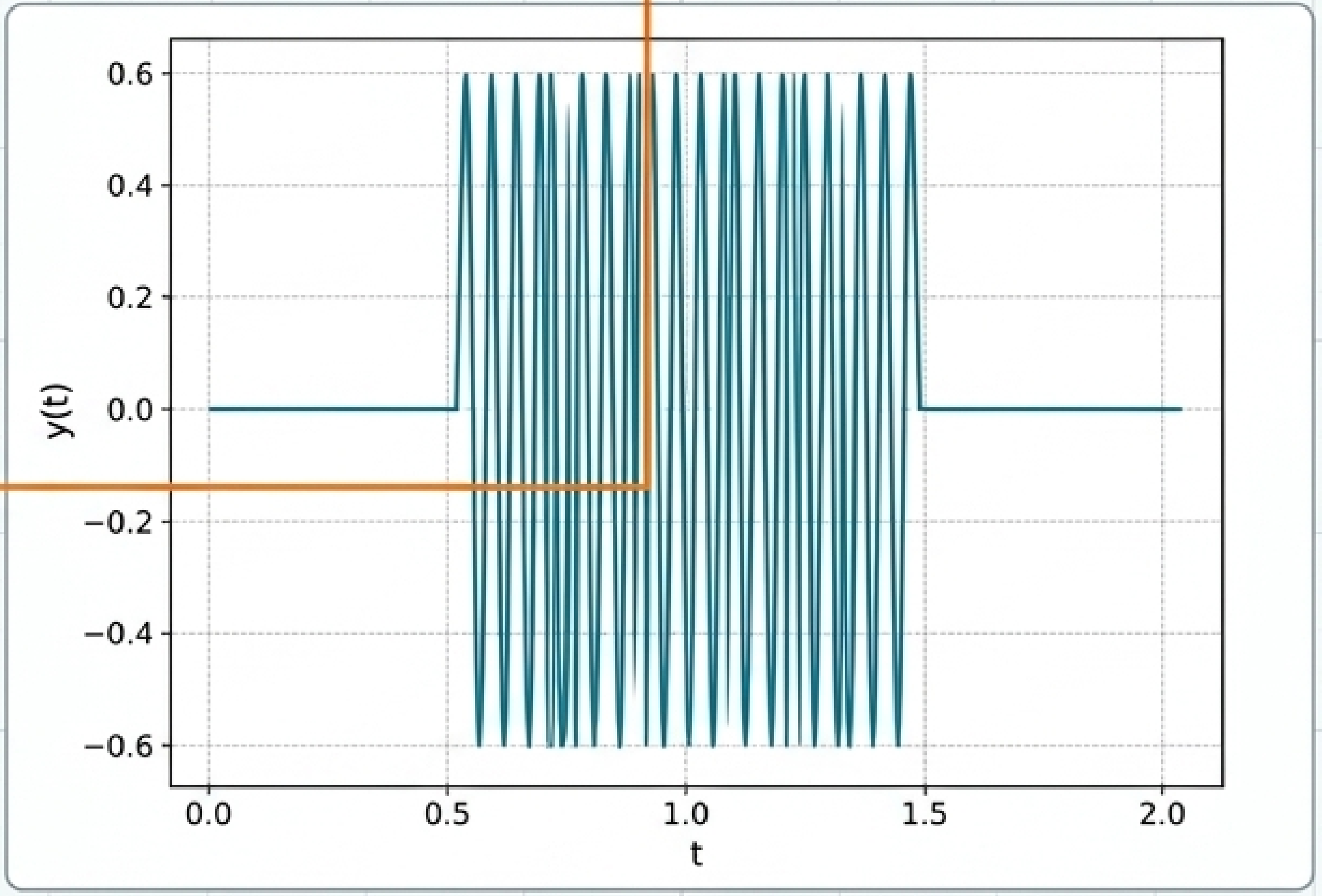
$$y(t) = 3 \cos(20\mathrm{pi}(t - 1) + \mathrm{pi}/4) * \mathrm{rect}(t)$$

Amplitude Scaling

A = 3

Frequency

omega_0 = 20pi -> f_0 = 10 Hz
(Period T_0 = 0.1s)



Time Transformation

Shifting the time base (t-1) means the signal is delayed by 1 second.

Classification

Continuous-time, Deterministic, Energy Signal (finite duration due to window), Real-valued.

Any complex real-world signal can be perfectly deconstructed using these foundational parameters.

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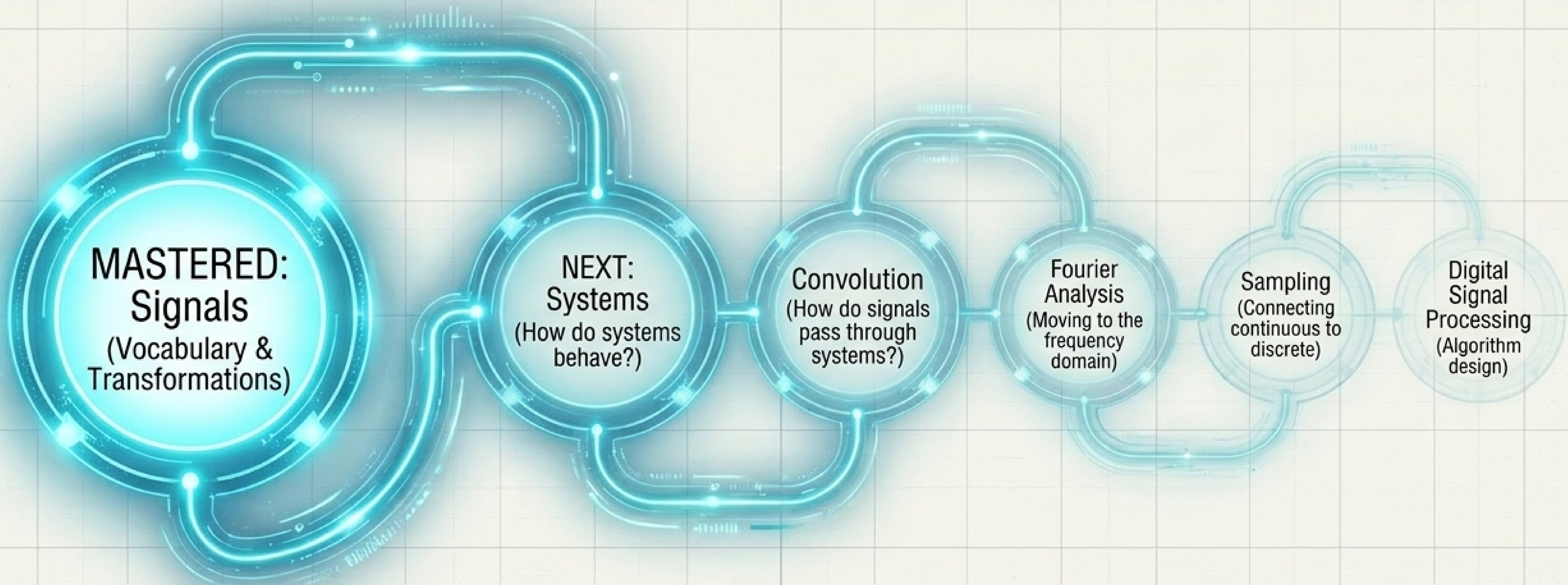

A strong command of definitions and transformations makes the mathematical development of the future stages substantially easier. The foundation is set.

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